COSC 31093 - Enterprise Software Design and Architecture

Project Assignment

Project Title: Airport Operations Management System

Group 15

Group Members:

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1. Overview

For our enterprise application, we chose to build an Airport Operations Management System, a platform that manages the day-to-day operations of a busy international airport. Running an airport involves a lot of moving parts: flights need to be scheduled, passengers need to be checked in, bags need to be tracked, security must stay on top of things, and staff and equipment need to be in the right place at the right time. A system that ties all this together needs to be fast, reliable, and able to handle thousands of events happening at once.

To build this properly, we designed this system using Microservices Architecture. Rather than building one massive application, we split the system into six focused services that each handle a specific part of airport operations. They run independently, talk to each other through APIs and event-driven messaging, and can be scaled or updated without touching the rest of the system. This makes the whole thing much easier to maintain and lets us scale specific parts like check-in when things get busy without wasting resources elsewhere.

2. Main Functional Areas

The system covers five core areas of airport operations:

- **Flight Operations -** Handles everything related to scheduling flights, managing arrivals and departures, assigning gates, and coordinating runway slots. The goal here is to reduce delays and keep airline timelines on track.
- **Passenger Processing -** Manages the passenger journey from the moment they check in to when they board the plane. This includes both counter check-in and self-service kiosks, boarding gate management, and immigration processing.
- **Baggage Handling & Tracking -** Tracks bags from check-in through to loading on the aircraft and retrieval at the destination carousel. This helps catch mishandling early and gives staff instant visibility of where any bag is at any point.
- **Security & Surveillance -** Covers monitoring of terminal areas, restricted zones, and passenger flow. It connects with camera systems and detection tools so security teams can respond quickly when something comes up.
- **Resource Management -** Handles the scheduling and allocation of staff, gates, vehicles, and ground equipment. It makes sure resources aren't double-booked and that everything needed for an operation is available.

3. Microservices Architecture

3.1 Service Decomposition

The six microservices below each own a distinct part of the system and are designed to be independently deployable:

Service	Functional Area	Primary Users	Key Capability
Flight Scheduling	Flight Operations	Airline Staff, Admins	Schedule & Conflict Management
Passenger Check-in	Passenger Processing	Passengers, Airline Staff	Check-in & Boarding
Baggage Tracking	Baggage Handling	Ground Staff	Real-time Bag Tracking
Security Monitoring	Security & Surveillance	Security Personnel	Surveillance & Alerts
Resource Management	Resource Operations	Admins, Ground Staff	Staff/ Gate/ Equipment Management
Notification & Alerts	Cross-cutting	All Users	Real-time Notifications

3.2 Why These Services Are Required

Each service was separated for a specific reason not just for the sake of it:

- **Flight Scheduling Service -** Airports deal with hundreds of flights a day, and schedules change constantly. Having a dedicated service means it can handle real-time updates and push changes to gates and ground crews without any dependency on the passenger systems.
- **Passenger Check-in Service -** Check-in volumes spike massively during peak hours. Keeping this as its own service means we can scale it up on demand without scaling up the entire system unnecessarily.
- **Baggage Tracking Service -** Lost or mishandled baggage is a serious operational problem. A dedicated tracking service lets us log every scan and transfer event independently, which makes it easy to trace exactly where a bag went wrong.
- **Security Monitoring Service -** Security can't afford downtime. By isolating it, we make sure that a slowdown in check-in or any other part of the system doesn't affect surveillance or alerting in any way.

- **Resource Management Service -** Allocating staff, gates, and equipment involves some complex scheduling logic. Separating this out means we can plug in optimization tools or third-party workforce systems without disrupting everything else.
- **Notification & Alert Service -** Every other service needs to send notifications to passengers, staff, or admins. Rather than building that logic into each service separately, we centralized it here, so it stays consistent and maintainable.

4. Intended Users

The system is designed to be used by five different groups, each with access to the parts relevant to them:

- **Airport Administrators -** Have full access to the system for monitoring operations, planning resources, and generating reports across all areas.
- **Airline Staff -** Use scheduling and check-in services to manage passenger manifests, handle boarding, and deal with any flight disruptions.
- **Security Personnel -** Work with the Security Monitoring Service to watch live feeds, respond to incidents, and keep logs of anything that happens.
- **Ground Handling Staff -** Use the baggage tracking and resource management services to coordinate luggage handling, equipment, and tarmac-side operations.
- **Passengers -** Interact with the system through self-service kiosks or the online portal to check in, get their boarding pass, check on their bags, and receive flight updates.

4. Reference(s)

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